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CDGCN: A CNN-enhanced dynamic graph convolutional network for multimodal fault diagnosis

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ABSTRACT

Intelligent fault diagnosis of rotating machinery under variable-speed and high-noise conditions remains challenge. Current deep learning-based approaches to multimodal fault diagnosis (e.g., using vibration and acoustic signals) face two major limitations. First, the approaches struggle to effectively fuse heterogeneous yet complementary information across different modalities. Second, when employing Graph Neural Networks (GNNs), they often rely on manually defined or static graph structures. This reliance prevents end-to-end, data-driven optimization and limits the adaptability to varying working conditions. To address these limitations, this paper proposes a novel CNN-enhanced Dynamic Graph Convolutional Network (CDGCN) for cross-modal fault diagnosis using vibration and acoustic signals. The CDGCN introduces an adaptively dynamic, data-driven graph construction strategy. A shared CNN first extracts local discriminative features from raw signals and maps them into probability distributions, from which Sinkhorn-approximated Wasserstein distances are computed to construct a dynamic adjacency matrix. The resulting graph structure enables a Graph Convolutional Network (GCN) to adaptively model inter-segment and cross-modal interactions, with the original CNN features serving as rich node attributes. This architecture effectively integrates temporal pattern extraction and spatial relationship learning, significantly enhancing diagnostic robustness under complex, variable-speed operating conditions. Extensive experiments on a self-developed multi-sensor bearing dataset validate the superiority of the proposed method, with classification accuracy exceeding 98.7% across various rotational speeds. Ablation and sensitivity analyses further confirm the effectiveness of dynamic graph construction and the critical role of the entropic regularization parameter ε.

1. Introduction

In modern industrial systems, mechanical equipment is undergoing continuous upgrades in complexity and performance. With the rapid development of advanced manufacturing technologies, there is a growing emphasis on the operational stability and safety of rotating machinery. Rolling bearings, as essential components within these systems, play a pivotal role in supporting rotational motion and load transfer. Their health status has a direct impact on the overall reliability and safety of industrial operations. As core elements in electromechanical systems, bearing failures can lead to unexpected downtime, substantial economic losses, and severe equipment damage. Therefore, achieving accurate and reliable health monitoring of bearings is of critical importance for ensuring safe and efficient system operation [1–4].

To meet the growing demand for accurate and real-time fault diagnosis, deep learning has emerged as a powerful tool due to its ability to automatically extract meaningful features from raw sensor data. Various neural network architectures—such as convolutional neural networks (CNNs) [5,6], recurrent neural networks (RNNs) [7,8], autoencoders (AEs) [9,10], long short-term memory networks (LSTMs) [11,12], and transformers—have been successfully applied in machinery fault diagnosis. These models outperform traditional methods by capturing complex temporal dependencies, spatial patterns, and structural relationships in vibration and acoustic signals, reducing reliance on manual feature engineering [13–15]. However, a significant practical challenge remains that industrial rotating machinery often operates under variable-speed conditions with high noise interference. Under such complex and non-stationary environments, the fault signatures captured by a single sensing modality (e.g., vibration) often become obscured, unreliable, and insufficient to fully characterize the machinery health states. This inherent limitation of single-modality data

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necessitates the fusion of complementary information from multiple sensing sources (e.g., vibration and acoustic) to enhance diagnostic robustness. Yet, despite this clear need, most existing deep learning models are still designed for and applied to single-modality data, lacking the inherent capability to effectively perform such heterogeneous fusion.

To address the multimodal fusion challenge, research efforts have explored fusion strategies at data, feature, and decision levels. For instance, Ma et al. [16] introduced a deep coupling autoencoder for multimodal signal fusion, while Ying et al. [17] developed a feature-enhance fusion network for acoustic and infrared data. Wan et al. [18] and Fu et al. [19] proposed networks coupling multi-sensor information. Zhang et al. [20] and Feng et al. [21] employed tensor representation and multi-source heterogeneous data fusion, respectively. Despite promising results, a common limitation persists that these methods predominantly rely on handcrafted or manually designed fusion architectures. Such fixed strategies struggle to capture the complex, dynamic, and nonlinear relationships among heterogeneous signals that naturally evolve under variable-speed conditions, often leading to degraded robustness and adaptability in real-word applications.

Graph Neural Networks (GNNs) offer a flexible and natural framework for modeling structural dependencies among multi-sensor signals within a unified graph structure [22,23]. By operating on graph-structured data and updating nodes through message passing [24,25], GNNs can learn embeddings that integrate local features and global information. Among GNN variants, Graph Convolutional Networks (GCNs) [26–28] have gained prominence for their efficiency. Recent studies have leveraged GNNs for multimodal diagnosis by constructing graphs where nodes represent sensors/modalities and edges encode similarities. For instance, Wang er al. [29] proposed a temporal-spatial graph neural network with an attention-aware module for fault diagnosis, constructing graphs by treating sensors as nodes, signal segments as attributes, selecting neighbors via cosine similarity and calculating edge weights with Gaussian kernel. Sun et al. [30] proposed a multi-sensor temporal-spatial graph network fused with empirical mode decomposition convolution for fault diagnosis, which constructs graph to mine spatio-temporal correlations. Meng et al. [31] introduced a multi-sensor fusion based on modal analysis and graph attention network for bearing fault diagnosis, which constructs graph structures via modal analysis (fusion video-derived phase vibration and acoustic signals, using singular value decomposition and frequency-domain decomposition to extract singular value vectors as nodes). Ren et al. [32] presented a cross-attention-based dual-branch graph neural network for multi-sensor fault diagnosis, which constructs graphs by taking sensors as nodes, judging edge existence through Pearson correlation coefficient and calculating edge weights via Gaussian kernel weight function. Li et al. [33] proposed an adversarially regularized graph wavelet autoencoder for unsupervised mechanical fault diagnosis by transforming multi-sensor signal into graph-structured representations and using spectral graph wavelet convolutions with adversarial regularization to extract multiscale features. However, these approaches introduce a critical limitation that their graph structures are typically predefined or constructed using fixed, non-learnable similarity metrics, resulting in a static topology that is decoupled from the downstream diagnostic task.

In summary, the existing multimodal bearing fault diagnosis methods are subject to three significant limitations:

1) Static or fixed multimodal fusion schemes: Many current methods, while capable of automatic feature extraction, adopt static fusion mechanisms—such as simple concatenation, weighted averaging, or pre-defined weights—that remain unchanged across different operating conditions. These strategies lack the adaptability to capture the evolving and context-dependent relationships between heterogeneous signals (e.g., vibration and acoustic), which often leads to suboptimal feature integration and reduced robustness under dynamic working conditions.
2) Handcrafted graph construction in GNNs: Although GNNs offers a structed way to model interactions among multimodal data, most GNN-based diagnostic models rely on predefined graph topologies. These are typically constructed using prior knowledge (e.g., sensor physical layout) or simple similarity metrics (e.g., cosine or correlation coefficients). Such graphs remain static throughout training and cannot adaptively reflect the latent, data-driven relationships between signal segments or across modalities, thereby limiting the ability to generalize under varying working conditions.
3) Limited data-driven dynamic interaction modeling: While some recent studies have explored adaptive fusion, there remains a scarcity of methods that fully integrate feature learning with dynamic structure learning in an end-to-end trainable framework. Most existing pipelines decouple representation extraction from relation modeling, which hinders the joint optimization of multimodal features and their interactive patterns, ultimately constraining diagnostic accuracy and system adaptability in real industrial environments.

This manual graph construction hinders fully data-driven, end-to-end training and reduces the adaptability and generalization capability of the models. To address the limitations, this paper proposed a novel CNN-enhanced Dynamic Graph Convolution Network (CDGCN) framework for multimodal fault diagnosis under variable-speed conditions. The main contributions of this paper are as follows:

1) A novel CNN-enhanced Dynamic Graph Convolutional Network (CDGCN) is proposed for multimodal fault diagnosis under variable-speed conditions. The framework jointly optimizes a shared CNN-based feature extractor and a dynamic graph convolutional network, where the graph topology is constructed in a fully data-driven and differentiable scheme employing Sinkhorn-approximated Wasserstein distance to derive a segment-adaptive adjacency matrix from distributional feature similarity. This design enables effective modeling of evolving intra-modal and cross-modal relationships without relying on static or predefined graph structures.
2) We establish a new, challenging self-developed multimodal bearing fault dataset under variable-speed conditions. Extensive comparative experiments are conducted against state-of-the-art methods, demonstrating that the proposed CDGCN achieves superior diagnostic accuracy (exceeding 99%) and robustness. Furthermore, detailed ablation and sensitivity analysis are performed to rigorously validate the effectiveness of the novel dynamic graph construction mechanism and the critical role of the entropic regularization parameter ε.

The remainder of the paper is organized as follows: Section 2 introduces the methodology of the.

proposed novel CDGCN. Section 3, provides the overview of the proposed method. Section 4 reports the experimental verification, and Section 5 presents ablation study. Finally, Section 6 summarizes the conclusions.

2. Methodology

2.1. CNN-based feature extraction

To capture discriminative temporal patterns from raw one-dimensional sensor signals, a lightweight Convolutional Neural Network (CNN) is employed as the front-end feature extractor. CNNs are particularly well-suited for feature extraction due to the ability to automatically learn hierarchical spatial–temporal representations through local receptive fields, and weight sharing, without relying on

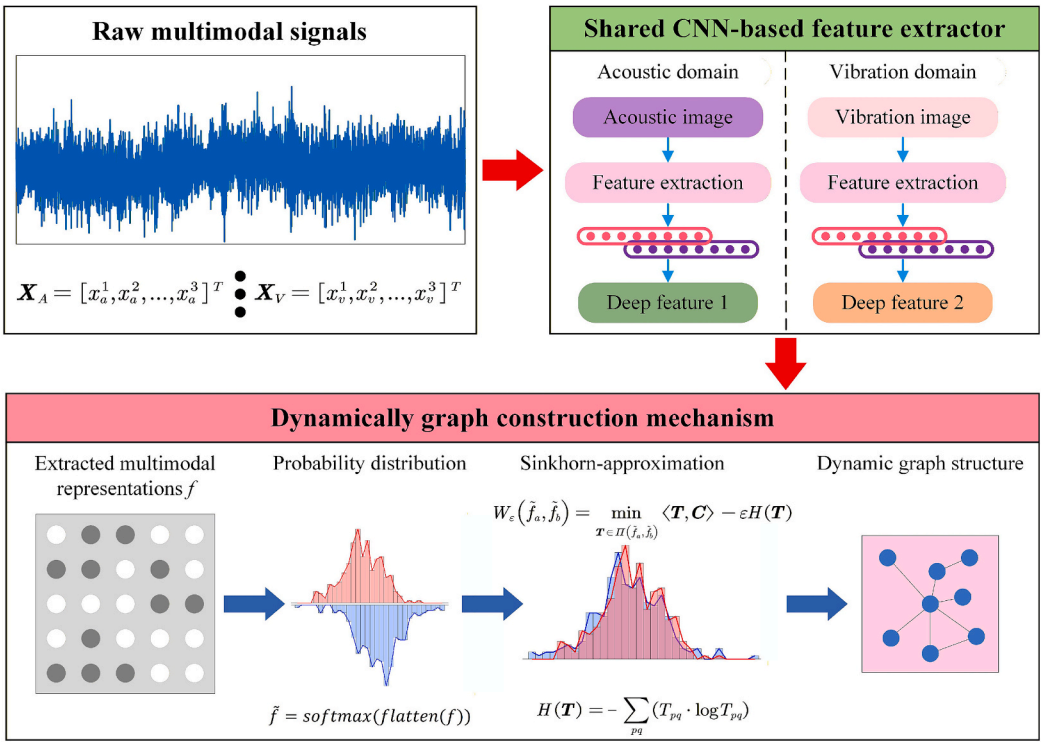

Fig. 1. Flowchart of dynamic graph construction process.

manual feature engineering. In this paper, each sensor node—originating from either the vibration or acoustic modality—provides a time-domain signal segment of fixed length, denoted as $x_i \in \mathbb{R}^{1\times L}$, where L represents the length of the time series, and the subscript i indexed the sensor node. In general, the output of a one-dimension convolution layer for the c-th output channel at time step t can be calculated as follow:

$$y_c(t) = \sum_{i=1}^{C_{in}} \sum_{j=0}^{K-1} w_{c,i,j} \cdot x_i(t+j) + b_c \tag{1}$$

where C_{in} denotes the number of input channels (typically 1 for one-dimension single-channel signals), K is the kernel size, $w_{c,i,j}$ and b_c represent the learnable convolution weight and bias term, respectively. After each convolution operation, a non-linear activation function $\sigma(\bullet)$ is applied to introduce non-linearity into the model, which is essential for learning complex patterns. In this paper, the ReLU is employed as the activation function due to its computational efficiency and effectiveness in mitigating the vanishing gradient problem. The output of the activation for the c-th channel at time step t is given by:

$$z_c(t) = \sigma(y_c(t)) = ReLU(y_c(t)) = max(0, y_c(t)) \tag{2}$$

where $z_c(t)$ is the output of the activation function for channel c at time step t; $y_c(t)$ denotes the output of the convolution operation at time step t. A pooling operation is employed to reduce the temporal resolution and increase robustness of the model. Unlike fixed-size pooling, an adaptive pooling layer is applied to dynamically downsample the temporal dimension to a predefined length, regardless of the input size. Specifically, the adaptive pooling outputs a feature map of fixed temporal length T', independent of the input length T. The output for channel c after adaptive pooling is:

$$f_c(t') = \frac{1}{|\mathscr{R}(t')|} \sum_{t \in \mathscr{R}(t')} z_c(t), t' = 1, 2, ..., T' \tag{3}$$

where $\mathscr{R}(t')$ denotes the set of indices in the input corresponding to the output position t'.

The final output of the CNN-based feature extractor for each sensor node is a two-dimension tensor $f_i \in \mathbb{R}^{d\times T'}$, where d is the number of convolutional output channels. These features are later interpreted as a distribution for computing inter node distances in graph construction. All sensor nodes share the same CNN weights, ensuring consistent feature representation across different modalities and reducing the number of trainable parameters.

2.2. *Wasserstein distance-based adaptive dynamic graph construction via Sinkhorn approximation*

An adaptive dynamic graph structure is constructed based on the distribution distance between node-level features, to model the latent interactions among cross-modal multi-sensor nodes, as shown in Fig. 1. To quantify the difference between nodes, the Wasserstein distance is adopted as a metric to evaluate the similarity between feature distributions. The Wasserstein distance provides a geometrically meaningful measure of the discrepancy between two probability distributions, which is especially suitable for comparing structured representations like feature maps. To enable end-to-end optimization, the Sinkhorn algorithm—a differentiable approximation to the original Wasserstein distance—is employed, which introduces an entropic regularization term for computational efficiency and gradient stability.

Given two node features $f_a, f_b \in \mathbb{R}^{d\times T'}$, probability distributions are obtained by applying the softmax function to their flattened representations, and can be expressed as follows:

$$\tilde{f}_a = softmax(flatten(f_a)) \tag{4}$$

$$\tilde{f}_b = softmax(flatten(f_b)) \tag{5}$$

where $\tilde{f}_a, \tilde{f}_b \in \mathbb{R}^{1\times dT'}$

The Sinkhorn distance is introduced as a differentiable approximation to the original Wasserstein distance, enabling end-to-end optimization of the entire model. Comparation with the traditional standard Wasserstein distance formulation, which is non-differentiable and

Table 1
The detailed calculation process of the Sinkhorn distance.

Algorithm 1 Sinkhorn distance calculation process	
Input:	Normalized feature vectors $\widetilde{f}_a, \widetilde{f}_b$ Regularization coefficient ε Maximum number of iterations N
1:	Construct cost matrix $\boldsymbol{C}$ using $C_{pq} = \left\| \widetilde{f}_a^{(p)} - \widetilde{f}_b^{(q)} \right\|_2^2$
2:	Compute kernel matrix $\boldsymbol{K} \in \mathbb{R}^{T' \times T'}$ using $\boldsymbol{K} = \exp\left(\frac{C}{\varepsilon}\right)$
3:	Initialize $\boldsymbol{U} = \boldsymbol{I}, \boldsymbol{V} = \boldsymbol{I}$
4:	for $t = 1$ to N do:
5:	$\boldsymbol{U}_{new} = \frac{\widetilde{f}_a}{\boldsymbol{K} \bullet \boldsymbol{V}}$ and $\boldsymbol{V}_{new} = \frac{\widetilde{f}_b}{\boldsymbol{K} \bullet \boldsymbol{U}}$
6:	if $\|\boldsymbol{U}_{new} - \boldsymbol{U}\| < \delta$ and $\|\boldsymbol{V}_{new} - \mathrm{V}\| < \delta$ do:
7:	break
8:	end if
9:	$\boldsymbol{U} = \boldsymbol{U}_{new}$ and $\boldsymbol{V} = \boldsymbol{V}_{new}$
10:	end for
11:	Compute approximate solution of transport matrix $\boldsymbol{T} = diag(\boldsymbol{U}) \bullet \boldsymbol{K} \bullet diag(\boldsymbol{V})$
12:	Compute Sinkhorn distance
13:	$W_\varepsilon\left(\widetilde{f}_a, \widetilde{f}_b\right) = \langle \boldsymbol{T}, \boldsymbol{C} \rangle = \sum_{p=1}^{T'} \sum_{q=1}^{T'} T_{pq} \bullet C_{pq}$
14:	return $W_\varepsilon\left(\widetilde{f}_a, \widetilde{f}_b\right)$

computationally intensive, the Sinkhorn algorithm incorporates an entropic regularization term, ensuring both smoothness and efficiency. Gradient information can be back-propagated from the graph construction module to the CNN-based feature extractor, adopting the Sinkhorn-based formulation, and the formulation can be expressed as follows:

$$C_{pq} = \left\| \widetilde{f}_a^{(p)} - \widetilde{f}_b^{(q)} \right\|_2^2 \tag{6}$$

$$\boldsymbol{C} = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1T'} \\ C_{21} & C_{22} & \cdots & C_{2T'} \\ \vdots & \vdots & \vdots & \vdots \\ C_{T'1} & C_{T'2} & \cdots & C_{T'T'} \end{bmatrix} \tag{7}$$

where $\boldsymbol{C} \in \mathbb{R}^{dT' \times dT'}$ denotes cost matrix, and here $d = 1$, C_{pq} represents the squared Euclidean distance between the feature vectors at temporal index p and q from nodes a and b, respectively. The Sinkhorn distance can be calculated by following formular:

$$W_\varepsilon\left(\widetilde{f}_a, \widetilde{f}_b\right) = \min_{\mathbf{T} \in \Pi\left(\widetilde{f}_a, \widetilde{f}_b\right)} \langle \mathbf{T}, \mathbf{C} \rangle - \varepsilon H(\mathbf{T}), \varepsilon > 0 \tag{8}$$

$$\Pi\left(\widetilde{f}_a, \widetilde{f}_b\right) = \left\{ \boldsymbol{T} \in \mathbb{R}^{T' \times T'} \middle| \boldsymbol{T} \bullet \boldsymbol{I} = \widetilde{f}_a, \boldsymbol{T}^T \bullet \boldsymbol{I} = \widetilde{f}_b, T_{ij} > 0 \right\} \tag{9}$$

$$H(\mathbf{T}) = -\sum\nolimits_{pq} \left(T_{pq} \bullet log T_{pq}\right) \tag{10}$$

$$\boldsymbol{I}^T = [1, 1, \cdots, 1] \tag{11}$$

where $\boldsymbol{T}$ denotes the transport matrix, $\Pi\left(\widetilde{f}_a, \widetilde{f}_b\right)$ represents the set of all joint probability distributions whose marginals match the given input $\widetilde{f}_a$ and $\widetilde{f}_b$, $H(\mathbf{T})$ is transport entropy, ε is the regularization coefficient, and the process for calculating the Sinkhorn distance is shown in Table 1.

To ensure that the dynamically constructed graph accurately reflects the underlying relationships among signal segments, three guiding principles are adopted during the graph construction process. These guidelines are designed to optimize node connectivity and enhance the representational power of the resulting graph structure.

(1) Node connection criterion: For each node in one modality (vibration or acoustic), the Sinkhorn distance between the node and the other four nodes. The connections between nodes are driven by the smallest distance, ensuring that each node is connected bidirectionally with the node most similar to it, improving the flow of information between nodes.
(2) Representative node selection: After establishing the initial bidirectionally connections, representative node in one domain is selected by calculating the number of edges each node is connected to. The representative node is the one with the most connections, effectively representing the main feature of its domain. This criterion ensures that key nodes in each modality are effectively represented, thereby improving the efficiency in handling multimodal data.
(3) Cross-modal connection: Finally, representative nodes of two modalities are connected, ensuring the strongest connection between vibration and acoustic modalities. This allows for more efficient information transfer between nodes of different modalities, thereby enhancing the ability to learn across modalities features.

To explicitly encode reliable intra-modal similarity and compact cross-modal correspondence while avoiding an overly dense topology, the dynamic graph is constructed under three guiding principles. These principles ensure that the constructed adjacency not only reflects sample-wise feature relationships, but also remains sparse, stable, and information-efficient for feature propagation.

(1) Intra-modal mutual-nearest connectivity principle.

For each node within a modality (vibration or acoustic), the Sinkhorn distance is calculated to all other nodes in the same modality and connect it to node with the minimum distance (the nearest neighbor under Sinkhorn metric), and can be expressed as follows:

$$j^* = \underset{j \neq i}{\operatorname{argmin}} W(i, j) \tag{12}$$

where j^* denotes the index of the nearest-neighbor node to i under the Sinkhorn distance, i and j denote the two distinct nodes within the same modality, respectively. $W(i, j)$ denote the Sinkhorn distance between the two nodes. A bidirectional edge between i and j^* is established. This mutual-nearest strategy yields a sparse but informative topology, preventing the graph from degenerating into a fully connected structure while still preserving the strongest local relations that are most beneficial for information propagation.

(2) Representative-node selection principle.

After the initial intra-modal connections are formed, a representative node was identified for each modality based on connectivity. Specifically, the number of established edges of the node computed and representative node of each modality was selective by follows:

$$deg^m(i) = \sum_{j=1, j \pm i}^{N_m} {A_{ij}}^m \tag{13}$$

$$r_m = \arg\left[\max_{i \in \{1, \cdots, N_m\}} deg^m(i) \right] \tag{14}$$

where, m denotes the modality index, and $m \in \{vibration, acoustic\}$, and N_m is the number of nodes in modality m, and in this paper$N_m = 5$. ${A_{ij}}^m$ represents the edge value of intra-modal adjacency matrix from node i to node j within modality m. $deg^m(i)$ is the number of neighbors connected to node i in modality m, and r_m denote the representative node of the modality m.

Table 2
The detailed dynamic graph structure construction process.

Algorithm 2 Dynamic graph structure construction process	
Input:	Vibration nodes $\boldsymbol{V} = [v_1, v_2, v_3, v_4, v_5]$, acoustic nodes $\boldsymbol{A} = [a_1, a_2, a_3, a_4, a_5]$, Node number N_{va}
Output:	Adjacency matrix $\boldsymbol{A}_{adj}$
1:	Initialize input feature matrix $\boldsymbol{M} = [\boldsymbol{V}, \boldsymbol{A}]$
2:	for $i = 1$ to N_{va} do:
3	Calculate Sinkhorn distance between node i and other four nodes in same modality
4:	Connect node i with the smallest Sinkhorn distance node in same modality bidirectionally
5:	end for
6:	Calculate the connection number of each node
7:	Choose the most connections one node to be the representative node in each modality
8:	Connect the representative nodes bidirectionally
9:	Save connection relationship between nodes as $\boldsymbol{A}_{adj}$
10:	return $\boldsymbol{A}_{adj}$

The representative node serves as a modality-specific prototype, encapsulating the canonical feature patterns inherent within its modality, thereby providing a robust and condensed representation for cross-modal interaction, which enhances fusion efficiency. This identification process is fully data-driven and adapts dynamically to the feature distribution of each input.

(3) Cross-modal connectivity principle.

Finally, cross-modal connection is constructed by linking the representative vibration node r_v and the representative acoustic node r_a. The edge weight is governed by their Sinkhorn distance, so that the cross-modal connection explicitly reflects the strongest and most reliable correspondence between modalities. By employing representative nodes rather than dense all-to-all cross-modal edges, the fusion becomes more targeted and less susceptible to noisy or weakly related node pairs, thereby enabling efficient and stable cross-modal information propagation.

By following these guidelines, dynamic graph structure (as detailed in Table 2) can dynamically adjust node connections and topology, ensuring that the model can automatically optimize and adapt to different input data patterns during training. Fig. 2 illustrates a visualized representation of the dynamically constructed graph, where the nodes correspond to the feature segments extracted from the vibration

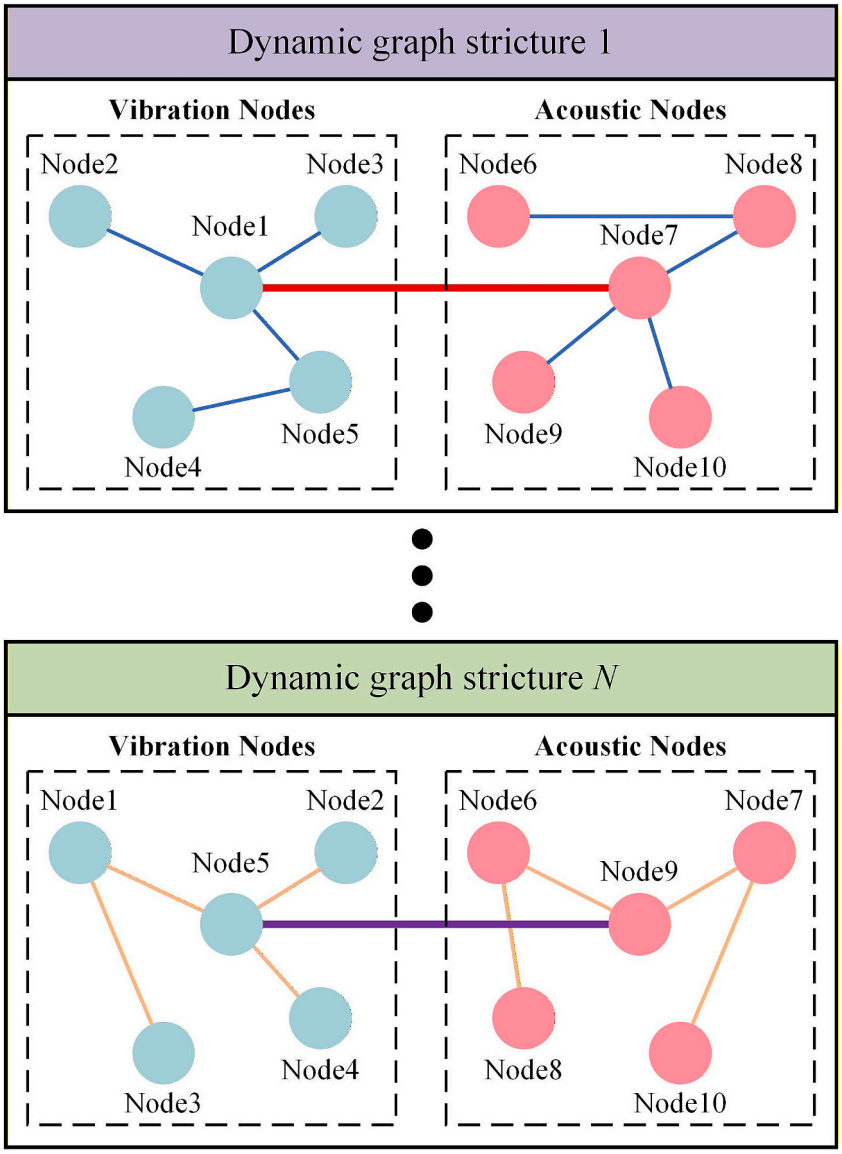

Fig. 2. Visualization of the dynamically constructed graph structure.

and acoustic modalities. It is essential to note that the visualization presents only an example of the graph topology, and in practice, the adjacency matrix and node connections are regenerated adaptively for every input instance based on the Sinkhorn-approximated Wasserstein distance-based dynamic construction mechanism.

2.3. Graph convolutional network

Graph convolutional network (GCN) is a type of neural network specially designed for graph-structured data, extending traditional convolution operations to non-Euclidean domains by aggregating feature information from each node's neighborhood. Once the adaptive graph is constructed, GCN is applied to capture the interactions between nodes within the dynamic graph structure.

Let $G = (V, E)$ denotes an undirected graph, where V is the set of N nodes, and E is the set of edges. The adjacency matrix $A \in \mathbb{R}^{N\times N}$ describes node connectivity, and the degree matrix $D \in \mathbb{R}^{N\times N}$ is a diagonal matrix with $D_{ii} = \sum_j A_{ij}$. The modified adjacency matrix is defined as $\widetilde{A} = A + I$ to ensure self-connections, where I is the identity matrix. Given an input node feature matrix $X \in \mathbb{R}^{N\times F}$, where F denotes number of features of each node, the propagation rule of a GCN layer is expressed as follows:

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}H^{(l)}W^{(l)}\right) \tag{15}$$

where $H^{(0)} = X$, $W^{(l)}$ is a learnable weight matrix of l layer, and D denotes the degree matrix. For graph classification, a graph-level representation from the node-level representations produced by the GCN layers is obtained by introducing global mean pooling, and the operation is as follows:

$$h_g = \frac{1}{N_{va}}\sum_{i=1}^{N_{va}} H_i \tag{16}$$

where H_i is the feature vector of the i-th node. Global mean pooling aggregate node features into a single vector that represents the entire graph. The graph-level representation h_g is passed through a fully connected layer W_f to output the final class label for the graph, and then a softmax function is applied to obtain prediction:

$$\widehat{y}_c = \frac{e^{\left(W_f h_g + b\right)_c}}{\sum_{i=1}^{M} e^{\left(W_f h_g + b\right)_i}} \tag{17}$$

here, $\widehat{y}_c$ is the predicted probability for class c, b is the bias term. The cross-entropy loss is introduced as loss function, and it can be described as follows:

$$\mathscr{L} = -\sum_{i=1}^{M} y_c \log(\widehat{y}_c) \tag{18}$$

where y_c is an indicator for the c-th class, if class c is the ground truth label $y_c = 1$, else $y_c = 0$. The Adaptive Moment Estimation (Adam) optimizer is applied as the model optimizer, and the express is as follows:

$$g_t = \nabla_\theta \mathscr{L}_t \tag{19}$$

$$m_t = \beta_1 \cdot m_{t-1} + (1-\beta_1)\cdot g_t \tag{20}$$

$$v_t = \beta_2 \cdot v_{t-1} + (1-\beta_2)\cdot g_t^2 \tag{21}$$

$$\begin{cases} \widehat{m}_t = \dfrac{m_t}{1-\beta_1^t} \\ \widehat{v}_t = \dfrac{v_t}{1-\beta_2^t} \end{cases} \tag{22}$$

Table 3
Detailed configuration of the proposed CDGCN.

Type of layer		Output size
Input		[1]
Conv-1	Conv 16@[1 × 3] × 1, stride = 1.BN, ReLU.	[16]
Conv-2	Conv 32@[1 × 3] × 1, stride = 1.BN , ReLU.Adaptive Pooling (size = 2), Linear-1:64 → 32. Softmax.Wasserstein distance calculated (via Sinkhorn approximation, $\varepsilon = 0.2$).Dynamic adjacency matrix (graph structure).	[32]
GCN-1	GCN 64@[32 × 64] × 1, BN, ReLU.	[640,64]
GCN-2	GCN 32@[64 × 32] × 1, BN, ReLUGlobal Pooling (Nodes = 10)	[32]
Classifier	Linear-2:64 → 16, Linear-3: 16 → 5.Softmax.	[5]

$$\theta_{t+1} = \theta_t - \alpha\cdot\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon} \tag{23}$$

where θ_t denotes the parameters of the network in the t iteration. $\nabla_\theta(\bullet)$ denotes the gradient of $(\bullet)$ with respect of θ. m_t, v_t is the gradient first moment estimate and the gradient second moment estimate in the t iteration respectively. α, ϵ is the learning rate and numerical stability term respectively.

3. Overview of the proposed CDGCN

To enhance fault diagnosis under cross-modal and variable conditions, a novel CNN-enhance Dynamic Graph Convolutional Network (CDGCN) is proposed, as summarized in Table 3. As presented in Fig. 3, CDGCN leverages two core components, a CNN-based feature extractor and a GCN module with a data-driven, dynamically constructed graph structure. Unlike conventional multimodal approaches that employ static or handcrafted fusion strategies, CDGCN introduces a fundamentally different paradigm in which the graph topology is not predefined but is adaptively inferred from the intrinsic statistical properties of the input multimodal signals.

The lightweight CNN-based feature extractor is designed to learn deep and discriminative representations from raw cross-modal signals (vibration and acoustic signals), effectively capturing localized spatial–temporal features across different sensing sources. To improve computational efficiency and ensuring consistency across nodes, all signal segments share the same feature extractor. This parameter-sharing strategy reduces redundant computations and ensures that all node features are embedded in a unified representation space, which is essential for stable graph construction and graph-based learning.

The key innovation lies in the subsequent adaptive graph construction module. Instead of relying on fixed similarity measures or sensor locations, the features, which are extracted from font-end CNN-based extractor, are first transformed into probability distributions using a Softmax-based normalization process. These probabilistic representations are then used to compute pairwise Sinkhorn-approximated Wasserstein distances between signal segments within the same modality. The Sinkhorn-approximated Wasserstein distance is chosen not only for its ability to measure distributional similarity in a geometrically meaningful way but also because it is differentiable, enabling end-to-end training of the entire framework. To convert the resulting distance matrix into a sparse yet informative topology, three guiding principles is introduced for dynamic construction. Specially, nodes are linked to their intra-modal nearest neighbors under the Sinkhorn distance with bidirectional edges, a representative node is identified in each modality by maximizing node connectivity, and the cross-modal interaction is established by connecting the representative nodes cross modalities. These principles prevent noisy dense coupling while preserving the most reliable relations for information propagation. Consequently, this

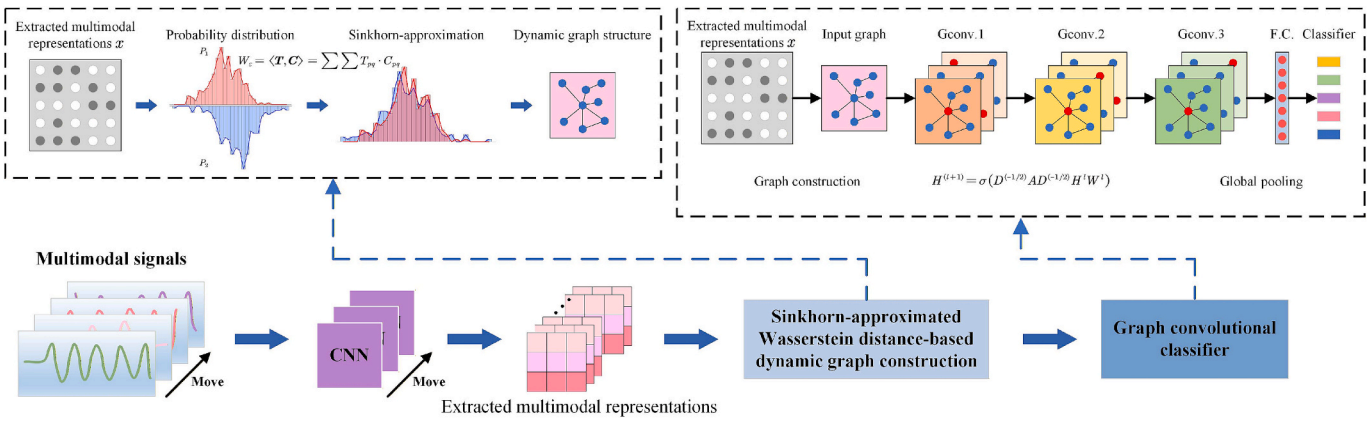

Fig. 3. Framework of CDGCN.

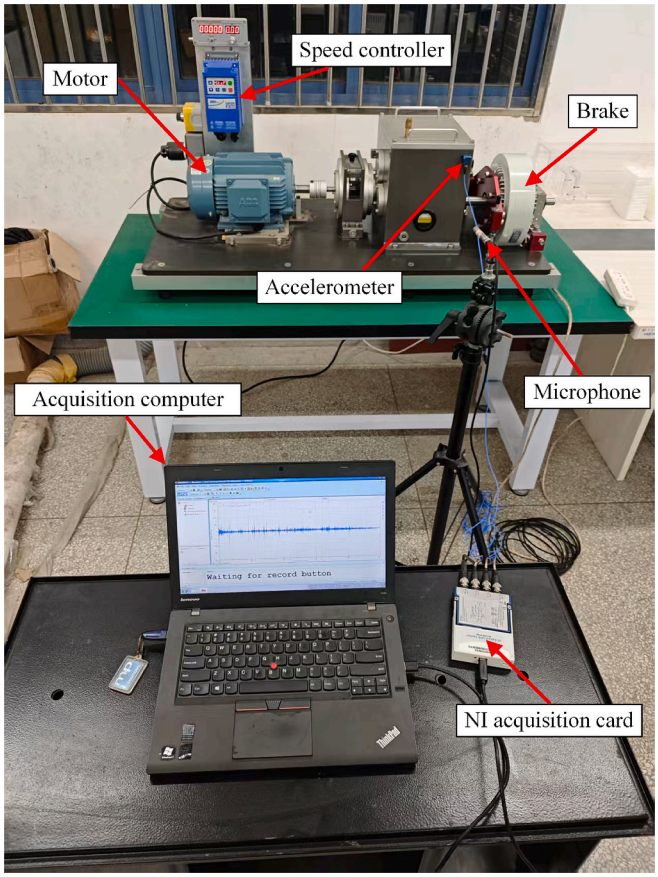

Fig. 4. Self-built platform.

procedure generates a condition-sensitive, data-driven dynamic adjacency matrix, which accurately reflects the evolving inter-node relationships influenced by varying operational speeds and cross-modal interactions.

Meanwhile, the original extracted features are retained and employing as node attributes into the GCN, where graph convolutional operations are performed based on the adaptively and differentially generated topology. By integrating distributional similarities for adaptive topology construction and semantically rich features for graph learning, the CDGCN framework effectively bridges cross-modal discrepancies and enhances model robustness.

4. Experimental verification

4.1. Experimental design

The effectiveness of the proposed method is verified through a self-built data acquisition platform (as shown in Fig. 4). The self-built platform is mainly composed of a motor, speed controller, gearbox, brake, five microphones and five accelerometers. There is a variable rotation speed of the motor (1200 rpm, 1800 rpm, 2400 rpm and 3000 rpm), and the brake is set to the released state. There are 0.5 mm deep and 1 mm wide grooves in the inner race and outer ring of the bearing, and 0.5 mm deep and 0.5 mm wide groove in the ball. Data from five health conditions of bearing is collected, including normal, inner race fault (IF), outer race fault (OF), rolling ball fault (BF) and hybrid fault (HF).

The sampling frequency of the microphones and accelerometers are all set equal to 12.8KHz. Each block remains 0.64 s and the block size is 8,192. Meanwhile, 15 blocks are acquired and total time is 9.6 s. Total length for each signal is 122,880 for one bearing health condition, and the data collected from different modalities and nodes under the representative 3000 rpm working condition is shown in Fig. 5. 24,576 length segment (20% of the total signal length) was used as the test signal. Train signal and test signal are both divided into segments of length 512 with an overlap of 256. As a result, the total number of samples is 386 for the training dataset and 95 for the test dataset under each health state and rotating speed.

4.2. Impact of entropic regularization parameter on diagnosis performance

In this section, the influence of hyperparameter ε on fault diagnosis classification performance when approximating th Wasserstein distance was investigated. By taking node 1 and node 2 in vibration domain under 1200 rpm rotational speed as an example, the 512-length segments from two nodes were first applied to calculate probability distributions across 32 histogram bins (align with the output length of the CNN-based feature extractor), as shown in Fig. 6, and it shows the probability mass functions (PMFs) of two nodes, using histograms and line plots. The distributional similarities and differences between the two nodes are clearly visualized, serving as a basis for calculating distribution distance.

Both the exact Wasserstein distance and Sinkhorn distance (entropic approximation Wasserstein distance) were computed, quantifying the distribution differences between the two nodes. As shown in Fig. 7(a), Sinkhorn distance varies significantly with entropic regularization parameter ε, while, Wasserstein distance remains constant. Fig. 7(b) presented the absolute difference between the Sinkhorn and Wasserstein distance. It is evident that the approximation error becomes negligible when ε is within the range of approximately 10^{-2} to 10^{-1}. This

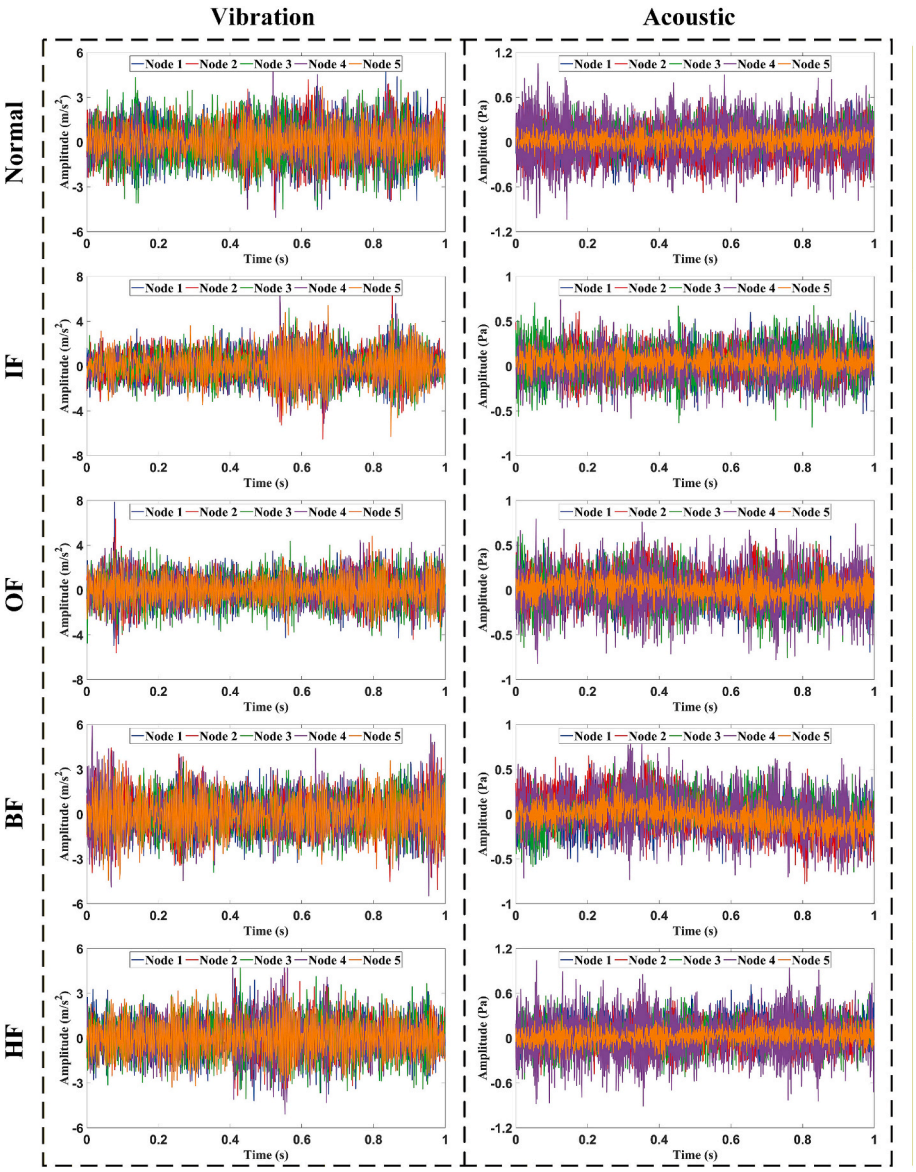

Fig. 5. Acquired experimental data under 3000 rpm rotational speed.

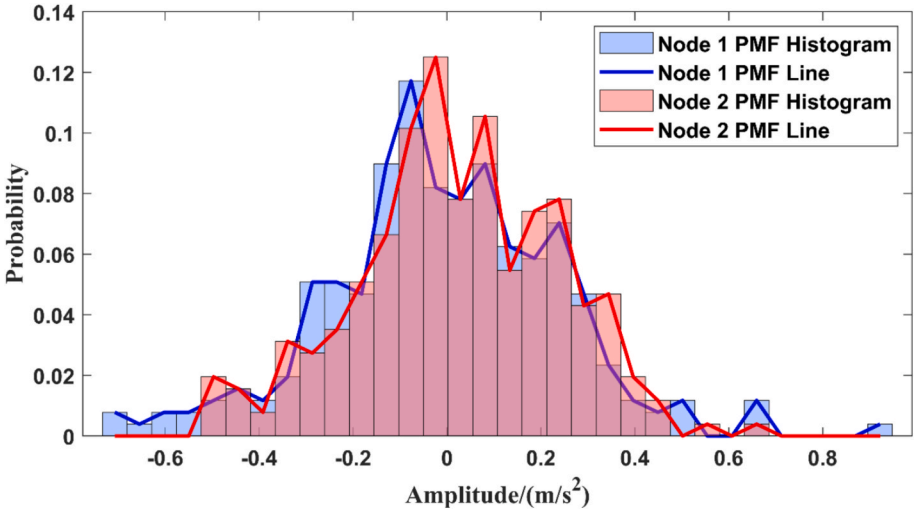

Fig. 6. Comparison of Probability Mass Functions.

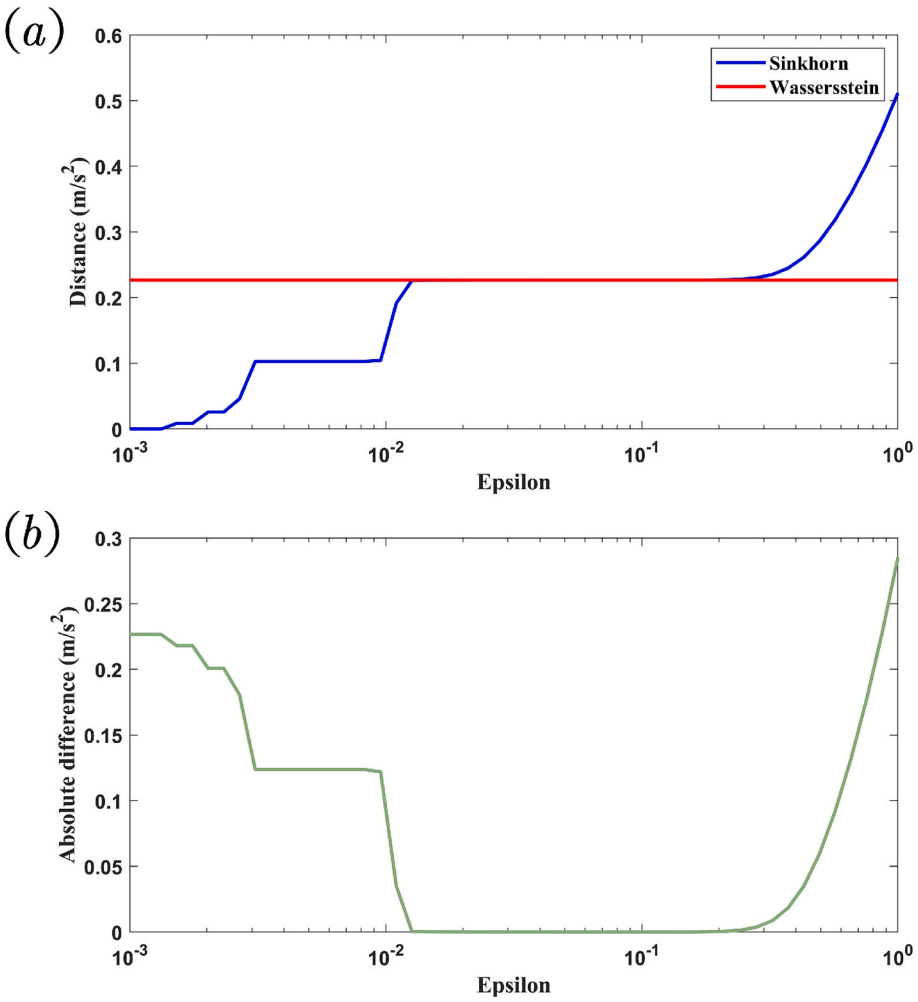

Fig. 7. Wasserstein and Sinkhorn distance differences *vs.* hyperparameter ε.

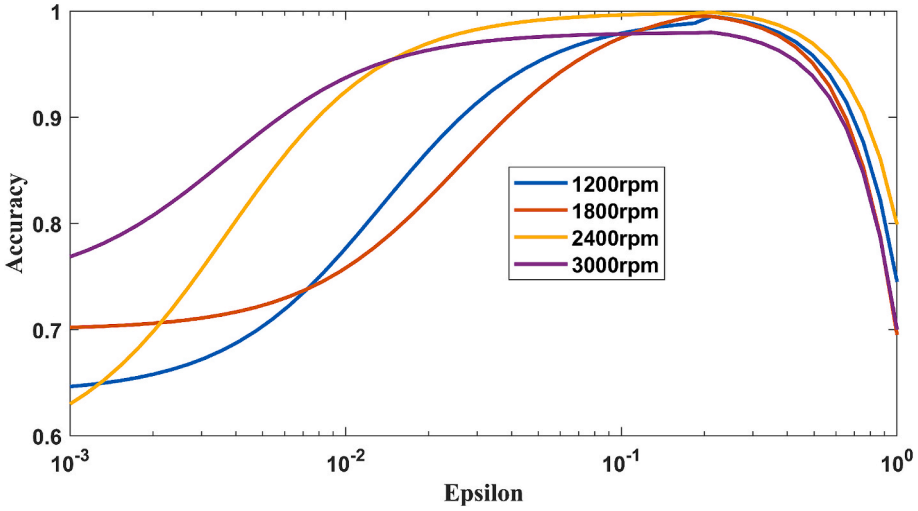

Fig. 8. Diagnostic accuracies *vs.* hyperparameter ε across under different conditions.

evaluation is not only intended to measure the distributional difference between node signals, but more importantly, the Sinkhorn distance is later employed to dynamically construct graph structure in a graph convolutional network (GCN). Therefore, selecting an appropriate ε is crucial to ensure both numerical stability and accurate structural representation within the dynamic graph.

Fig. 8 expressed the impact of the entropic regularization parameter ε on fault diagnosis accuracy under different rotation speed conditions. It is observed that the classification accuracy generally improves as ε increases from 10^{-3} to approximately 10^{-1}. However, a further increase

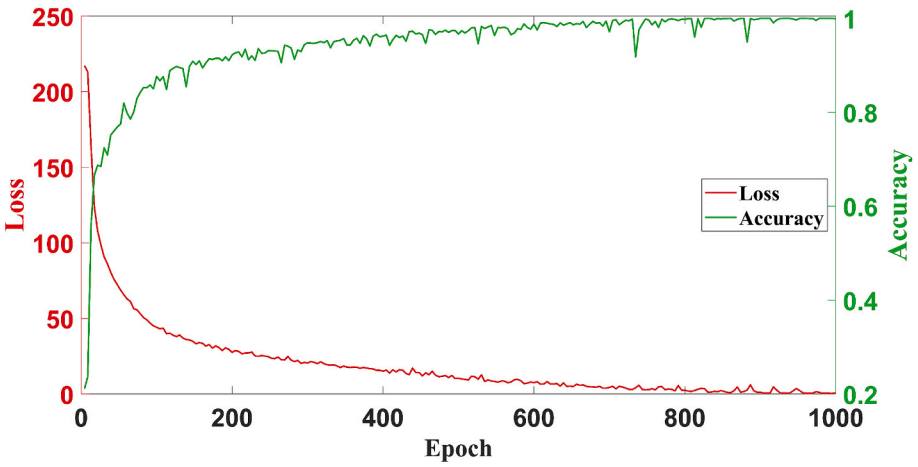

Fig. 9. Convergence of training process.

Table 4
Accuracy of fault diagnosis under different working conditions and test sequences [%].

Working condition	Test sequence					Average	STD
	Test 1	Test 2	Test 3	Test 4	Test 5		
1200 rpm	99.16	99.16	99.37	98.53	98.95	99.03	0.28
1800 rpm	99.37	99.16	98.53	98.95	99.37	99.08	0.31
2400 rpm	99.79	100.00	100.00	100.00	100.00	99.96	0.08
3000 rpm	98.74	98.95	99.16	98.32	98.53	98.74	0.30

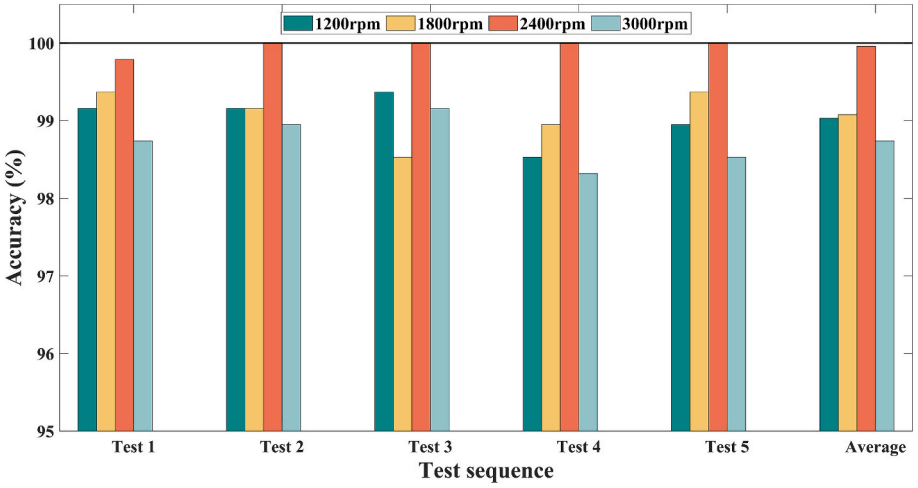

Fig. 10. Diagnostic results of five tests under different working conditions.

in ε beyond this range leads to a sharp decline in accuracy across all speed conditions. The performance peak for each speed occurs within the range of $\varepsilon = 0.1$ to 0.3, indicating that a moderate regularization strength facilitates more effective graph construction and fault feature learning in the proposed CNN-enhanced Dynamic Graph Convolution Network (CDGCN) model. Notably, the model achieved near-perfect accuracy when ε is appropriately selected, underscoring the robustness of the proposed CDGCN method, and in this paper ε is selected to 0.2.

4.3. Diagnosis Effect analysis

To verify the training stability and convergence of the proposed CNN-enhanced Dynamic Graph Convolution Network (CDGCN), the training curve under 3000 rpm working condition is presented in Fig. 9, which shows both the loss and accuracy metrics over 1000 epochs. As shown in Fig. 9, the loss curve demonstrates a rapid decrease in the initial epochs, indicating the high learning efficiency in capturing essential features from the multimodal data. After approximately 800 epochs, the loss curve stabilizes to a low value, suggesting that the model has effectively converged. Crucially, the accuracy curve exhibits smooth, consistent growth that closely tracks the performance improvement, plateauing near the maximum value without significant oscillations. This stability confirms the robust generalization capability of the CDGCN framework and validates the effectiveness of the Sinkhorn-approximated dynamic graph construction in maintaining a stable gradient flow, thereby avoiding overfitting and ensuring reliable learning across the multimodal signals.

The diagnostic accuracies of the proposed CDGCN under different rotation speeds and five independent test sequences, are illustrated in Table 4 and Fig. 10. Across all working conditions, the proposed method

Table 5
Fault diagnosis Accuracy of competitive methods under different conditions [%].

Method	Working condition				Average
	1200 rpm	1800 rpm	2400 rpm	3000 rpm	
DRMNN	72.35 ± 1.45	71.13 ± 0.70	70.36 ± 2.42	74.81 ± 0.66	72.16 ± 2.25
MDCFN	68.84 ± 1.52	70.11 ± 1.91	69.78 ± 2.34	71.54 ± 1.55	70.07 ± 2.10
GATU-Net	82.19 ± 1.76	80.51 ± 1.12	81.43 ± 2.02	82.95 ± 1.85	81.77 ± 1.94
MBGCN	82.20 ± 1.81	84.45 ± 0.63	83.80 ± 1.26	81.46 ± 1.38	82.98 ± 1.80
FIGNN	85.63 ± 1.41	83.92 ± 0.84	84.73 ± 1.31	82.67 ± 0.77	84.24 ± 1.56
GCMN	81.02 ± 1.71	81.87 ± 0.62	82.95 ± 1.19	81.63 ± 1.84	81.87 ± 1.59
DGNN	82.65 ± 1.29	83.96 ± 0.92	81.60 ± 1.49	82.11 ± 0.83	82.58 ± 1.46
Proposed method	99.03 ± 0.28	99.08 ± 0.31	99.96 ± 0.08	98.74 ± 0.30	99.20 ± 0.53

consistently achieved high diagnostic accuracy, with average accuracies of 99.03%, 99.08%, 99.96% and 98.74% at 1200 rpm, 1800 rpm, 2400 rpm and 3000 rpm, respectively. Notably, the remarkably low standard deviations (STD) of 0.28%, 0.31%, 0.08% and 0.30% corresponding to each speed further confirm the excellent repeatability and robustness of the method. Individual test accuracies remained consistently high, with the lowest recorded value still above 98%, and several test sequences achieving perfect 100% accuracy.

Table 5 and Fig. 11 presented a comprehensive comparison of fault diagnosis accuracy between the proposed method and six state-of-the-art methods under different working conditions, including: 1) DRMNN [19], which uses a dynamic routing algorithm to adaptively weigh and fuse feature from different modalities. 2) MDCFN [34], which is designed for cross-modal fault diagnosis by integrating multiple modalities. 3) GATU-Net [35], a graph attention-based network which constructs a graph by representing each sensor as a node, with edges established based on the cosine similarity between corresponding signal segments. 4) MBGCN [36], a model that integrates Gaussian filters and a multibranch GCN with enhanced multi-receptive fields to effectively diagnose faults. 5) FIGNN [37], a GCN guided by fuzzy inference, and 6) GCMN [38], a graph convolutional memory network to assess the whole-life defect evolution of rolling bearings. 7) DGNN [39], a dynamic graph neural network to constructs and updates graph structures employing Gaussian similarity for accurate fault diagnosis.

As comprehensively detailed in Table 5, the proposed CDGCN decisively outperformed all other advanced comparative methods, achieving a remarkable average accuracy of 99.20% with notably low variability (±0.53%). This performance significantly surpasses the suboptimal method, FIGNN, which had an average accuracy of 84.24% with standard deviations ranging from ± 0.77% to ± 1.41% across different speed conditions. Notably, compared to the dynamic graph construction method DGNN, which achieved an average accuracy of 82.58% (± 1.46%), our proposed CDGCN demonstrates a clear and substantial improvement, highlighting the effectiveness of the Sinkhorn-approximated Wasserstein distance in adaptively modeling multi-modal relationships under variable speeds. While other methods exhibited moderate accuracy with greater fluctuations, CDGCN consistently demonstrated both superior accuracy and exceptional stability across all speeds. Even the lowest accuracy (98.74% at 3000 rpm) exceeds the best average accuracy of any comparative method, underscoring the robustness and reliability of the proposed method under various operational conditions.

The confusion matrices under the four working conditions are shown in Fig. 12, and provides further evidences of the effectiveness of the proposed method. The results show minimal misclassification, especially for Normal and OF conditions, which are clearly differentiated by the model. While there is some misclassification between BF and HF health conditions, the overall performance remains strong, with high diagnostic accuracy across all health states and working conditions. Additionally, the Uniform Manifold Approximation and Projection (UMAP), as shown in Fig. 13, is introduced to further validate the discriminability of the features extracted by CDGCN across different bearing health states. In Fig. 13, the UMAP visualizations for the proposed method under four different rotational speeds are shown. These visualizations highlight that the features extracted by CDGCN are well-separated for different health states, demonstrating that the model can effectively distinguish between various fault types under different working conditions.

Furthermore, to comprehensively assess the practical deployment potential of CDGCN, the performance analysis with a critical evaluation of computational complexity was conducted. As quantitatively compared in Table 6, CDGCN demonstrates a decisive advantage in model efficiency over the three most competitive methods—MBGCN, FIGNN, and GCMN. Specifically, the proposed method contains only 9,125 parameters, which is approximately 80% fewer than MBGCN—the most parameter-efficient baseline among the three methods. This substantial reduction in model size is directly reflected in the inference speed. CDGCN requires merely 0.87 s for inference, operating 4.4 times faster than MBGCN, 5.3 times faster than FIGNN, and 8.3 times faster than GCMN. These results conclusively show that

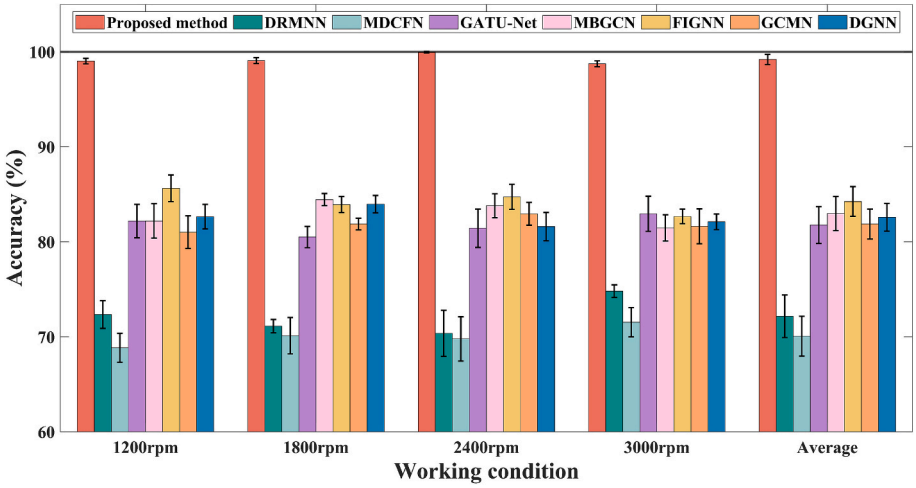

Fig. 11. Accuracy comparison with various state-of-the-art methods under different speeds.

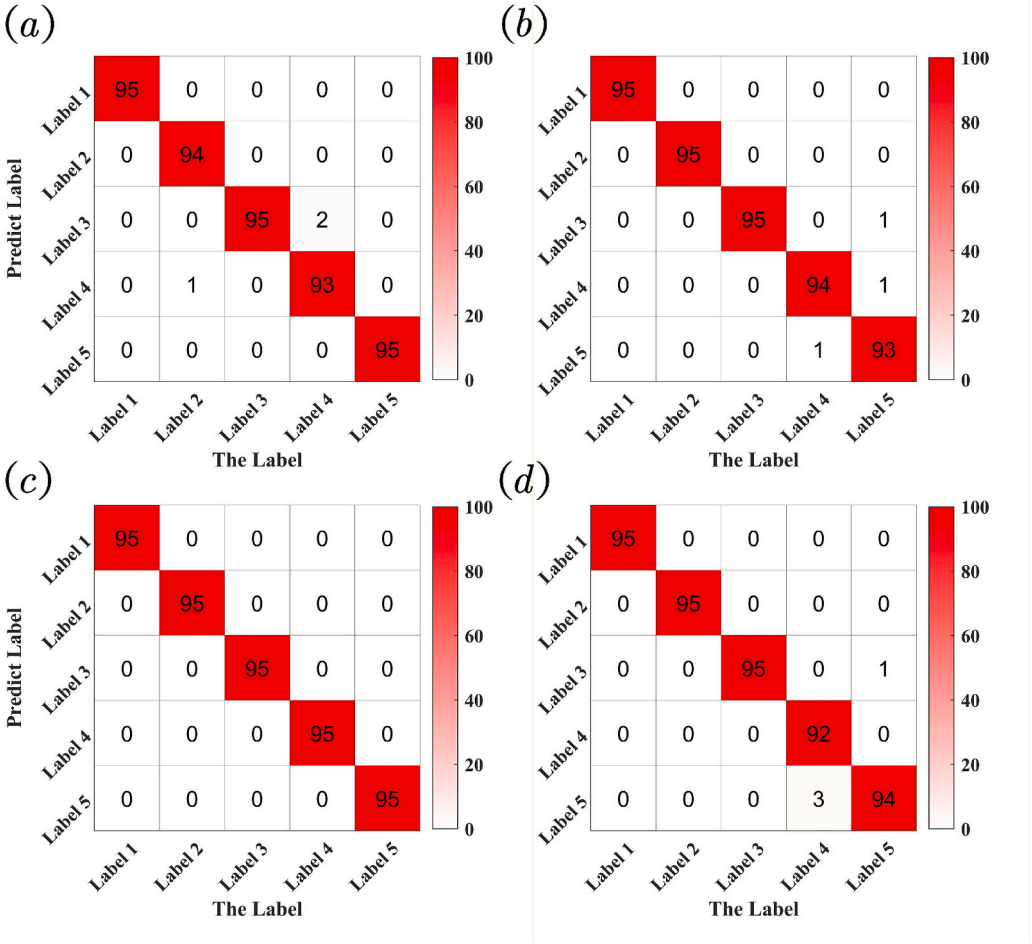

Fig. 12. Confusion matrix under different working conditions. (a) 1200 rpm, (b) 1800 rpm, (c) 2400 rpm, (d) 3000 rpm.

CDGCN achieves superior diagnostic accuracy not at the expense of computational resources, but through a more efficient architectural design. The significantly lower parameter counts and faster inference speed, coupled with top accuracy, make CDGCN exceptionally suitable for real-time industrial fault diagnosis scenarios where both high reliability and limited computational resources are critical concerns.

The collective evidence from quantitative metrics, detailed classification analysis, and feature distribution visualization solidly demonstrates that CDGCN establishes a new state-of-the-art in cross-modal fault diagnosis. The fundamental architectural and methodological advancements result these superior experimental outcomes. The seamless integration of a CNN-based feature extractor with a GCN creates a cohesive model that simultaneously captures localized temporal patterns and global relational structures. Crucial, the innovative, fully data-driven graph construction mechanism—which leverages Sinkhorn-approximation Wasserstein distances on learned feature distributions—allows the model to adaptively infer and encode the intrinsic relationships among multi-source signal segments without relying on any predefined topology. This end-to-end differentiable design enables the joint and synergistic optimization of both the node representations and the graph structure. It is this synergistic co-adaptation between feature learning and relational reasoning that endows CDGCN with its exceptional diagnostic accuracy, remarkable stability and strong generalization capability under various working conditions.

5. Ablation study

5.1. Effectiveness analysis of key modules

To further evaluate the contribution of key modules in the proposed framework, an ablation study was conducted by selectively removing different components, and three methods are constructed as follows:

(1) *Method 1*: The proposed CDGCN model.
(2) *Method 2*: The model is obtained by removing CNN-based feature extractor in the proposed method.
(3) *Method 3*: The model is obtained by removing Wasserstein distance-based dynamic graph structure in the proposed method, static graph structure instead.

Each experiment was conducted five time under different rotational speeds, with the average accuracy and standard deviation presented in Table 7. Method 1 (the completely proposed model) consistently achieved the highest diagnostic performance across all working conditions, with average accuracies of 99.03 ± 0.28%, 99.08 ± 0.31%, 99.96 ± 0.08%, and 98.74 ± 0.30% at 1200 rpm, 1800 rpm, 2400 rpm, and 3000 rpm, respectively. Removing the CNN-based feature extractor (Method 2) led to a pronounced decline in diagnostic accuracy (e.g., 74.53 ± 1.93% at 1200 rpm) demonstrating that the deep discriminative features learned by the CNN are essential for reliable fault classification. Similarly, substituting the dynamic graph structure with a static topology (Method 3) also resulted in considerable performance degradation,

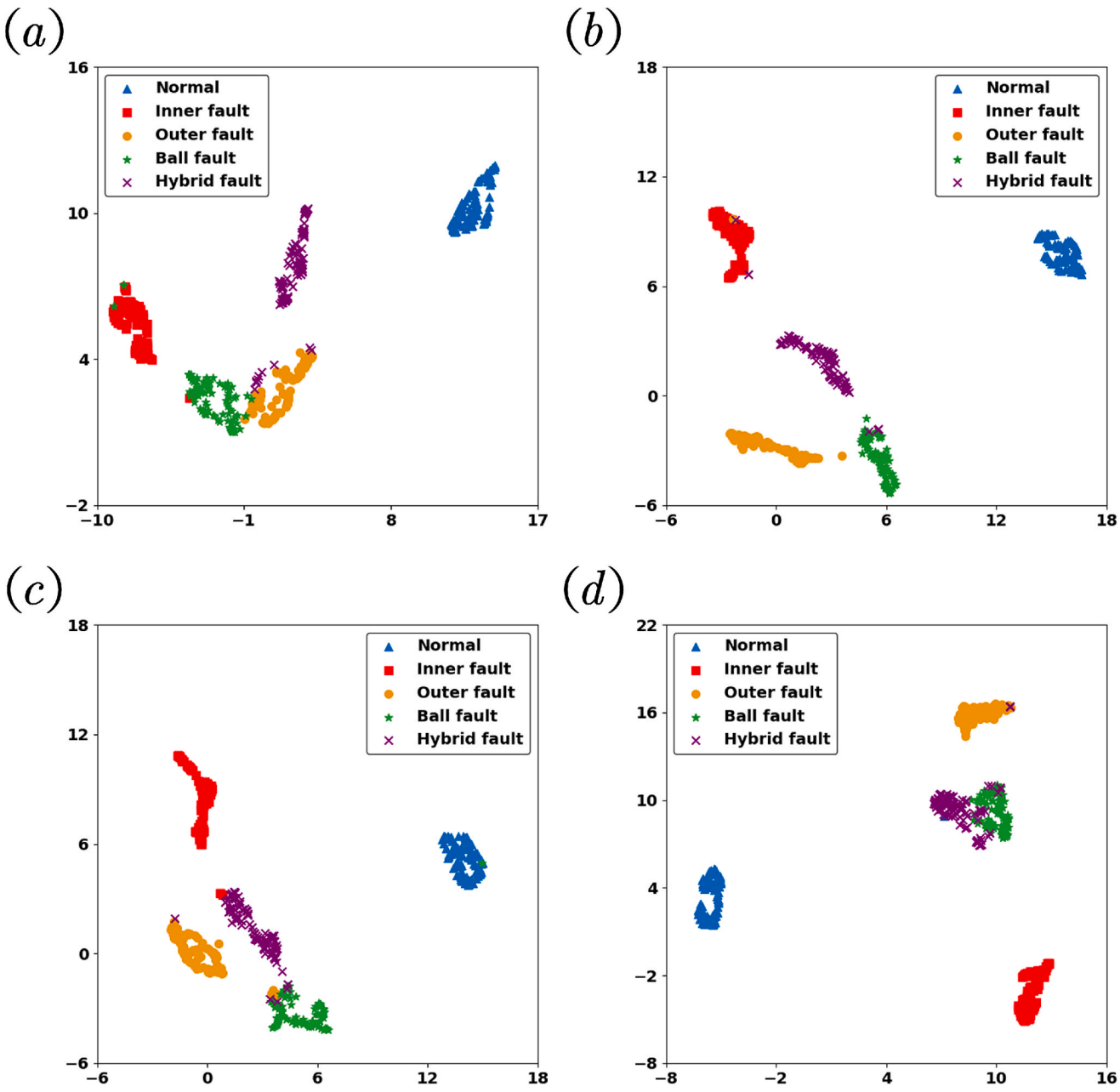

Fig. 13. UMAP visualization under different working conditions. (a) 1200 rpm, (b) 1800 rpm, (c) 2400 rpm, (d) 3000 rpm.

Table 6
Computational complexity comparison.

Method	Parameters	Inference Latency
MBGCN	47,461	3.84 s
FIGNN	62,129	4.63 s
GCMN	140,293	7.29 s
Proposed method	9,125	0.87 s

particularly under high rotational speeds (dropped to 76.34 ± 0.94% at 3000 rpm).

These results underscore that both the CNN-based feature extractor and the Wasserstein distance-driven dynamic graph construction are critical to the robustness and generalization of the proposed CDGCN framework.

Table 7
Diagnostic accuracies with different methods [%].

Method	Working condition				Average
	1200 rpm	1800 rpm	2400 rpm	3000 rpm	
Method 1	99.03 ± 0.28	99.08 ± 0.31	99.96 ± 0.08	98.74 ± 0.30	99.20 ± 0.53
Method 2	74.53 ± 1.93	76.06 ± 1.28	78.81 ± 2.08	74.47 ± 2.31	75.97 ± 2.62
Method 3	81.01 ± 1.99	83.92 ± 1.67	80.06 ± 2.97	76.34 ± 0.94	80.33 ± 3.38

5.2. Dynamic graph construction mechanism Validation

To quantitatively verify the effectiveness and superiority of our proposed dynamic graph construction mechanism based on the Sinkhorn-approximated Wasserstein distance compared to conventional

Table 8
Performance comparison of different graph construction mechanisms [%].

Method	Working condition				Average
	1200 rpm	1800 rpm	2400 rpm	3000 rpm	
Method 4	99.03 ± 0.28	99.08 ± 0.31	99.96 ± 0.08	98.74 ± 0.30	99.20 ± 0.53
Method 5	81.01 ± 1.99	83.92 ± 1.67	80.06 ± 2.97	76.34 ± 0.94	80.33 ± 3.38
Method 6	78.36 ± 2.18	80.17 ± 2.20	81.78 ± 1.10	79.68 ± 2.92	80.00 ± 2.52
Method 7	83.21 ± 1.36	84.43 ± 0.99	83.75 ± 2.36	81.96 ± 1.17	83.34 ± 1.80
Method 8	84.51 ± 1.47	85.01 ± 1.13	82.27 ± 1.26	83.45 ± 1.97	83.81 ± 1.83

approaches, a crucial ablation experiment within the CDGCN framework is conducted by replacing the original graph construction mechanism with other methods, and four methods are constructed as follows:

(1) *Method 4*: The proposed CDGCN model.
(2) *Method 5*: The model is obtained by replacing Sinkhorn-approximated Wasserstein distance-based graph mechanism with a static graph structure based on the physical location of sensors.
(3) *Method 6*: The model is obtained by replacing original graph construction mechanism in CDGCN framework with a fully connected topology structure where all nodes are bidirectionally connected.
(4) *Method 7*: The model is obtained by replacing original construction mechanism in CDGCN framework with a method that calculates the similarity between different nodes using cosine similarity and bidirectionally connects the most similar nodes.
(5) *Method 8*: The model is obtained by replacing original construction mechanism in CDGCN framework with a method that calculates the similarity between different nodes using Gaussian similarity and bidirectionally connects the most similar nodes.

The comparative results of the model performance when using these different graph construction mechanisms are summarized in Table 8, and Fig. 14. As presented in Table 8, and Fig. 14, the proposed Method 4 achieves the highest classification accuracy and the lowest standard deviation, demonstrating superior performance and stability. In contrast, Method 5 exhibits noticeably lower accuracy (e.g., 76.34 ± 0.94% at 3000 rpm) and higher variance, indicating that fixed sensor location-based graph structure lack adaptability under various working conditions. Method 6 (fully connected graph) performs still substantially worse than Method 4, with accuracies ranging from 78.36 ± 2.18% to 81.78 ± 1.10%, and shows greater fluctuations in 3000 rpm conditions. Notably, Method 6 even underperforms Method 5 (sensor location-based static graph), indicating that a fully connected topology, by forcing every node to aggregate features from all others equally, not only introduces noisy, non-discriminative information but also undermines the beneficial structural prior based on physical proximity and may accelerate feature over-smoothing in the model, ultimately leading to degraded diagnostic performance. Method 7 achieves slightly better diagnostic accuracy than Method 5 and 6, yet average accuracy (83.34 ± 1.80) remains far below that of the proposed approach, underscoring the limitation of cosine similarity in capturing distributional relationships compared to the Sinkhorn-approximated Wasserstein distance. Furthermore, Method 8, which employs Gaussian similarity for dynamic edge construction, achieves an average accuracy of 83.81% (± 1.83%), performing comparably to Method 7 but still significantly lower than our proposed Method 4. This presentation suggests that while Gaussian-based similarity can capture local relationships in the feature space, it still falls short in capturing the nuanced distributional discrepancies between multimodal signal segments under variable-speed conditions, compared to the Wasserstein distanced-based mechanism.

These results collectively confirm that the proposed Sinkhorn-approximated Wasserstein distance-based dynamic graph construction mechanism is not only more accurate but also more robust and condition-adaptive than the conventional and advanced graph construction strategies, thereby validating the critical role in the CDGCN framework.

6. Conclusion

In this paper, a novel CNN-enhanced Dynamic Graph Convolutional Network (CDGCN) was proposed for cross-modal fault diagnosis under different working conditions. The framework successfully integrates a CNN-based feature extractor with a dynamically constructed graph convolutional network, where the graph topology is adaptively built using Sinkhorn-approximated Wasserstein distances to model inter-segment and cross-modal dependencies.

Extensive experiments on a self-built test platform demonstrated that the proposed CDGCN significantly outperforms several state-of-the-art methods, achieving robust diagnostic accuracies above 98.5% across all tested rotation speeds. Importantly, computational complexity analysis confirmed that the CDGCN maintains high computational efficiency and a practical inference time, ensuring viability for real-time industrial deployment. Ablation studies further confirmed the critical role of both the CNN-based temporal feature extractor and the

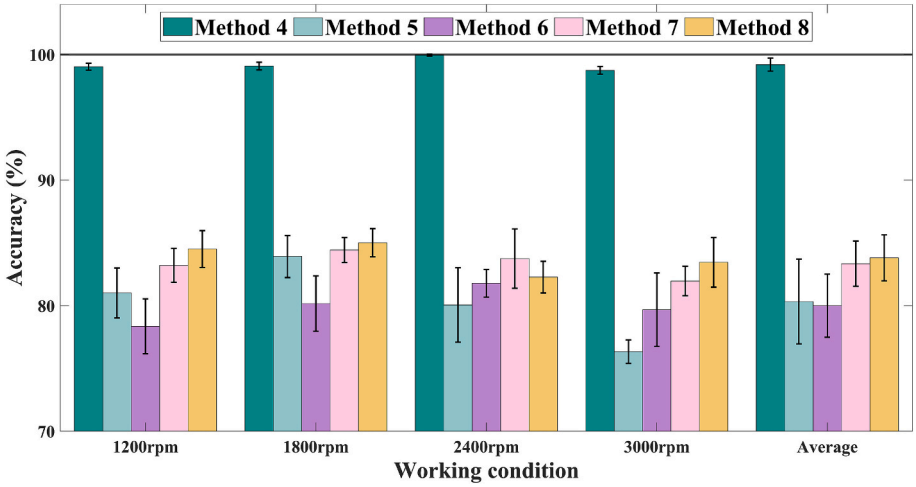

Fig. 14. Diagnostic accuracy with different graph construction mechanisms.

adaptively dynamic graph construction strategy in achieving the superior performance. Moreover, sensitivity analysis confirmed that the entropic regularization parameter $\varepsilon = 0.2$ yield optimal diagnostic results.

In conclusion, CDGCN provides a scalable and effective solution for multi-modal fault diagnosis in realistic industrial scenarios. Future research will focus on enhancing the model with attention-based graph mechanisms and extending its applicability to online and incremental diagnostic tasks with real-time data streams.

CRediT authorship contribution statement

Songhua Xiao: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Data curation. **Junqi Hou:** Visualization, Validation. **Muhammad Jamshaid Khan:** Visualization, Validation. **Beibei Sun:** Writing – original draft, Supervision, Resources, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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